Thm. $[a,b] \subset \mathbb{R}$ cpct.

Pf. assume seq. cpct.

(x_n) seq in $[a,b]$. Then $x := \limsup x_n = \lim_{n\to\infty} \sup_{m\geq n} x_m \in [a,b]$ as $\sup_{m\geq n} x_m \in [a,b]$

Given $k \in \mathbb{N}$, $\epsilon = \frac{1}{k}$, $\exists N$ s.t. $\left|x - \sup_{m\geq N} x_m\right| < \epsilon/2 \rightsquigarrow \exists n_k \geq N : |x - x_{n_k}| < \epsilon$.

i.e. $x \in \left(\sup x_m - \frac{\epsilon}{2}, \sup x_m + \frac{\epsilon}{2}\right)$

Indeed, $\exists n_k$ s.t.

$\left|x_{n_k} - \sup_{m\geq N} x_m\right| < \epsilon/2$

then $|x - x_{n_k}| < \epsilon$.

So $x_{n_k} \to x$.

$$\int_0^1 (1-x)^{np}\,dx \underset{\substack{u=1-x \\ du=-dx}}{=} \int_0^1 u^{np}\,du = \frac{1}{np+1}$$

$$\frac{1}{p} + \frac{1}{q} = 1$$

$$\frac{1}{q} = 1 - \frac{1}{p} = \frac{p-1}{p} \rightsquigarrow q = \frac{p}{p-1}$$

$f : \mathbb{R} \to \mathbb{R}$ cts, cpct supp. K Then $\lim_{p\to\infty} \|f\|_p = \|f\|_\infty$

$$\|f\|_p = \left(\int_K |f(x)|^p\right)^{1/p} \qquad \|f\|_\infty = \sup_{x\in K} |f(x)|.$$

want $\left|\left(\int_K |f(x)|^p\right)^{1/p} - \sup_{x\in K} |f(x)|\right| < \epsilon$.

$$\int_K |f(x)|^p\,dx \longrightarrow \sup |f(x)|^p \quad ?$$

$$\leq \int_K \sup|f(x)|$$

$$\frac{1}{1 + d(x,y)} \geq \frac{1}{d(x,z) + 1}$$

$- d(z,y)$.

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ℓ^∞ complete.

Proof. Let $(a^{(n)})$ be a Cauchy seq in ℓ^∞. That is, $\forall \epsilon>0\ \exists N\in\mathbb{N}$ st. $m,n>N \Rightarrow$

$$\|a^{(m)}-a^{(n)}\|_\infty = \sup_{k\in\mathbb{N}} |a_k^{(m)}-a_k^{(n)}| < \epsilon.$$

Given $k\in\mathbb{N}$, $(a_k^{(n)})_n$ is Cauchy seq in $\mathbb{R}$. Indeed $\forall\epsilon>0$ let N be as above,

then $|a_k^{(m)}-a_k^{(n)}| \le \sup|\ -\ | < \epsilon$.

by $\mathbb{R}$ complete, $a_k^{(n)} \longrightarrow a_k$.

Claim: $a=(a_k)_{k\in\mathbb{N}} \in \ell^\infty$ (1) and $a^{(n)} \longrightarrow a$ (2).

(1). Cauchy $\Rightarrow$ bdd so $\|a^{(n)}\|_\infty \le K\ \forall n \Rightarrow |a_k^{(n)}| \le \|a^{(n)}\| \le K\ \forall n \forall k$

so $\lim_{n\to\infty} |a_k^{(n)}| \le K$

$\quad\quad\quad

5 $f: \mathbb{R} \to \mathbb{R}$ cts, $\operatorname{supp}(f) \subset K$. Then $\lim_{p\to\infty} \|f\|_p = \|f\|_\infty$.

Proof. $\|f\|_p = \left(\int_{\mathbb{R}} |f(x)|^p \, dx\right)^{1/p} \leq \left(\int_K \|f\|_\infty^p \, dx\right)^{1/p} = \|f_\infty\| \underbrace{\left(\int_K 1 \, dx\right)^{1/p}}_{\|f_\infty\| \text{ as } p\to\infty.}$

So $\lim_{p\to\infty} \|f\|_p \leq \|f_\infty\|$

OTOH. $\sup_{x\in K} |f(x)|$ is attained, say at $x_0 \in K$.

Let $U \ni x_0$ be s.t. $|f(x)| > \|f\|_\infty - \epsilon$ on $x \in U$.

Then
$$\begin{aligned}
\|f\|_p &= \left(\int_K |f(x)|^p dx\right)^{\frac{1}{p}} \\
&\geq \left(\int_U |f(x)|^p dx\right)^{\frac{1}{p}} \\
&\geq \left(\int_U (\|f\|_\infty - \epsilon)^p dx\right)^{\frac{1}{p}} \\
&= (\|f\|_\infty - \epsilon)\left(\int_U 1 \, dx\right)^{\frac{1}{p}} \\
&\longrightarrow \|f\|_\infty - \epsilon.
\end{aligned}$$

So $\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty - \epsilon \longrightarrow \|f\|_\infty$

so $=$.

6. $p < q \Rightarrow \|a\|_p \geq \|a\|_q$.

Proof. wlog $\|a\|_p = 1 \Rightarrow \sum |a_n|^p$ so each $|a_n|^p \leq 1$

$\|a\|=0$ trivial.

$\Rightarrow |a_n| \leq 1$

$\Rightarrow p<q \Rightarrow |a_n|^q < |a_n|^p$.

$$\begin{aligned}
\|a\|_q &= \left(\sum_{n=1}^\infty |a_n|^q\right)^{\frac{1}{q}} \\
&\leq \left(\sum_{n=1}^\infty |a_n|^p\right)^{\frac{1}{q}} \\
&= \|a\|_p^{\frac{p}{q}} \\
&= 1 = \|a\|_p.
\end{aligned}$$

if $q = \infty$, then $\|a\|_\infty = \sup |a_n|$

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1. ($\Rightarrow$) $a_n \to a$. Let a_{n_k} be a subseq. Then $a_{n_k} \to a$. Indeed let $\epsilon > 0$.

Let $N \in \mathbb{N}$ be st. $|a_n - a| < \epsilon$ $n > N$.

Let $K \in \mathbb{N}$ be st. $n_k > N$ $k > K$

Then $|a_{n_k} - a| < \epsilon$ $k > K$. So every subseq $\to a$.
this is a subseq of itself.

($\Leftarrow$) $a_n \not\to a$. Then $\exists \epsilon > 0$ st. $\forall N \in \mathbb{N}$ $\exists n > N$ st. $|a_n - a| \geq \epsilon$.

Let $N = 1$. $\exists n_1$ s.t. $|a_{n_1} - a| \geq \epsilon$.

recursively, let $N = n_{k-1}$. $\exists n_k > n_{k-1}$ s.t. $|a_{n_k} - a| < \epsilon$.

Then (a_{n_k}) stays ϵ-away from a, so no subseq goes to a.

2. $X = C([0,1])$. $f_n(x) = x^n$. $p < \infty$.

$$\|f_n\|_p^p = \int_0^1 x^{np}\,dx = \frac{1}{np+1} \longrightarrow 0 \text{ as } n \to \infty.$$

3. $|F(f) - F(g)| = |f(0) - g(0)| \leq \|f - g\|_\infty$ So F cts in L^∞.

not in $p < \infty$: $f_n(x) = (1-x)^n$

$$\|f_n(x)\|_p^p = \int_0^1 (1-x)^{np}\,dx = \frac{1}{np+1} \longrightarrow \infty. \text{ But } F(f_n) = 1 \ \forall n.$$

4. ✓

5. $\operatorname{supp}(f) \subset K$.

$$\|f\|_p = \left(\int_{\mathbb{R}} |f(x)|^p\,dx\right)^{\frac{1}{p}}$$
$$\leq \left(\int_{\mathbb{R}} \|f\|_\infty^p\,dx\right)^{\frac{1}{p}}$$
$$= \|f\|_\infty \left(\int_{\mathbb{R}} dx\right)^{\frac{1}{p}} \xrightarrow{p \to \infty} \|f\|_\infty.$$

$\|f\|_\infty = \sup_{x \in K} |f(x)|$ is attained at some $x \in K$. By continuity, $\exists U$ st.

$|f(x)| \geq \|f\|_\infty - \epsilon$ on $x \in U$.

Then
$$\|f\|_p = \left(\int_K |f(x)|^p\,dx\right)^{\frac{1}{p}}$$
$$\geq \left(\int_U |f(x)|^p\,dx\right)^{\frac{1}{p}}$$
$$\geq \left(\int_U (\|f\|_\infty - \epsilon)^p\,dx\right)^{\frac{1}{p}}$$
$$= (\|f\|_\infty - \epsilon)\left(\int_U dx\right)^{\frac{1}{p}} \xrightarrow[\epsilon \to 0]{p \to \infty} \|f\|_\infty.$$

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6. $p \mapsto \|a\|_p$ dec in p

$p < q < \infty$. if $a_n = 0$ then $\|a\|_q \le \|a\|_p$.

else wlog $\|a\|_p = 1$. Then $|a_n|^p \le 1 \ \forall n$.

$p<q \Rightarrow |a_n|^q \le |a_n|^p$. $\qquad |a_n| \le 1 \quad \forall n$.

$$\|a\|_q = \left(\sum_{n=1}^{\infty} |a_n|^q\right)^{\frac{1}{q}} \le \left(\sum_{n=1}^{\infty} |a_n|^p\right)^{\frac{1}{q}} = \|a\|_p^{\frac{p}{q}} = 1. \quad ✓$$

if $q=\infty$, then

$$\|a\|_\infty = \sup|a_n| \le \left(\sum_{n=1}^{\infty} |a_n|^p\right)^{\frac{1}{p}} = \|a\|_p.$$

7. $$a_k^{(n)} = \begin{cases} \frac{1}{n^{\frac{1}{p}}} & k \le n \\ 0 & k > n \end{cases}$$

$$\|a^{(n)}\|_p = \left(n \cdot \frac{1}{n^{\frac{1}{p}\cdot p}}\right)^{\frac{1}{p}} = 1 \quad \forall n \not\to 0.$$

$$\|a^{(n)}\|_q = \left(n \cdot \frac{1}{n^{\frac{q}{p}}}\right)^{\frac{1}{q}} = \left(n^{\frac{1}{q}} \ n^{-\frac{1}{p}}\right) \longrightarrow 0 \text{ as } n \to \infty,$$

as $\frac{1}{q} - \frac{1}{p} < 0$.

$$\|a^{(n)}\|_\infty = \frac{1}{n^{\frac{1}{p}}} \to 0 \quad \text{as } n \to \infty.$$

8 ✓

9 ✓

10. $f(x) = \sin(2\pi x)$.

$f'(x) = 2\pi \cos(2\pi x)$

$f''(x) = -4\pi^2 \cdot \sin(2\pi x)$

$\vdots$

$$\|f^{(n)}\|_\infty = (2\pi)^n$$

So $$\sum_{n=0}^{\infty} 2^{-n} \|f^{(n)}\|_\infty = \sum \pi^n = \infty.$$

1. ℓ^∞ complete. $(a^{(n)})$ Cauchy. Then for each $k \in \mathbb{N}$, $|x_k^{(n)} - x_k^{(m)}| \le \|x^{(n)} - x^{(m)}\|_\infty$

$x \in \ell^\infty$: $\sup_k |x_k| \le \sup_n \|x^{(n)}\| < \infty$ as $x^{(n)}$ Cauchy. Cauchy in $\mathbb{R}$ so $\to x_k$. $x = (x_k)$.

$x^{(n)} \to x$: Given $\epsilon > 0$, N st. $m,n > N \Rightarrow \|x_m - x_n\|_\infty < \epsilon$. i.e. $|x_k^{(n)} - x_k^{(m)}| < \epsilon$

$\downarrow$

$|x_k - x_k^{(m)}| \le \epsilon \ \forall k, m > N$. So $\|x - x^{(m)}\| \le \epsilon$.

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2. $B \subset \ell^1$ $p \in [1,\infty]$ not cpct.

$\overset{\shortparallel}{\{(a_n) \in \ell^1 : \|a\|_p \le 1\}}$.

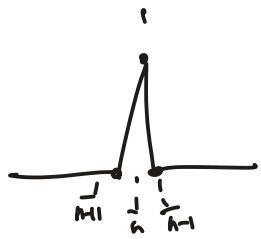
$$a_k^{(n)} = \begin{cases} 1 & k = n \\ 0 & k \neq n. \end{cases}$$

$\|a^{(n)}\|_p = 1 \quad \forall p, \forall n.$

but $\|a^{(n)} - a^{(m)}\|_p = 1$ for $n \neq m$ so not Cauchy $\rightsquigarrow$ cannot cges.

3. Clear.

4. $C([0,1])$ unit ball L^∞ not cpct.

$f_n(x) =$ 

$|m-n| \ge 2 \Rightarrow \|f_n - f_m\| = 1$ so no subsq is Cauchy.

P 2.13 $\quad p_n \to p \in M \Rightarrow f(p_n) \to q \in N.$

if f not cts at a, then $\exists\, \epsilon > 0$ st. $\forall \delta > 0$ $\exists |x-a| < \delta$, $|f(x) - f(a)| \ge \epsilon$.

$\delta = \frac{1}{n}$: $|x_n - a| < \delta$, $|f(x_n) - f(a)| \ge \epsilon$.

$x_1, a, x_2, a, \dots \rightsquigarrow f(x_1), f(a), f(x_2), f(a), \dots$

cges to a as $\forall \epsilon > 0$ $\frac{1}{n} < \epsilon \Rightarrow |x_n - a| < \frac{1}{n} < \epsilon$

so $\to a$.

$\Longrightarrow$ cges but subseq $f(a) \to f(a)$ while $f(x_i) \not\to f(a)$ ↯.

37. $f: \mathbb{R} \to \mathbb{R}$ cts only at $\mathbb{Z}$.

$$f(x) = \begin{cases} \sin(\pi x) & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q}. \end{cases}$$

19 is $\mathbb{Q}$ homeo $\mathbb{N}$? No, pts in $\mathbb{N}$ open but pts in $\mathbb{Q}$ not

22. Yes, M complete. $x_n \in M$ Cauchy. Thm. bded, so contained in some $\bar{B}(R)$ $\downarrow$ cmpct

So x_n has a subseq $\to x \in M$.

by Cauchy $x_n \to x$.

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43. cts image of cpct set is cpct. ETS π_1, π_2 cts.

46. A, B cpct disjnt. nonempty $\subset M$. $\exists a_0 \in A, b_0 \in A$ st.

$d(a_0, b_0) \leq d(a,b)$. $\forall a \in A, b \in B$.

ETS $d: A \times B \to \mathbb{R}$ cts. Then by EVT it attains min.

Given $\epsilon > 0$, $a, b \in A \times B$ wts $\exists \delta > 0$ st.

$d((a',b'),(a,b)) < \delta \Rightarrow |d(a',b') - d(a,b)| < \epsilon$.

$\| $

$d(a',a) + d(b',b) < \delta = \frac{\epsilon}{2}$ $\qquad d(a',b') \in (d(a,b) - \epsilon, d(a,b) + \epsilon)$.

$d(a',b') \leq d(a',a) + d(b',b) + d(a,b) < d(a,b) + \epsilon$.

$d(a',b') \geq d(a,b) - d(a',a) - d(b',b) > d(a,b) + \epsilon$.

52. (a) $A_1 \supset A_2 \supset \cdots$ nonempty closed sets of M complete.

$\operatorname{diam} A_n \to 0$. $\bigcap A_n$ is 1 pt.

(b) $\operatorname{diam} A_n \not\to 0$ for $[a, \infty)$.

(a). $a_i \in A_i$ then a_i is Cauchy why? $\operatorname{diam} A_n \to 0 \Rightarrow$

$\forall \epsilon > 0$, $\exists N$ st. $\operatorname{diam} A_n < \epsilon$. for $m, n < N$, $a_m, a_n \in A_m$ $m \geq n$ wlog.

$\Rightarrow |a_m - a_n| \leq \operatorname{diam} A_n < \epsilon$.

M complete $\Rightarrow$ $a_n \to a$. Claim $a \in A_i$ $\forall i$ Indeed

a_i is limit pt of A_i by tail. So $a \in \bigcap A_n$.

$\bigcap A_n \subset \{a\}$: if $b \neq a \in \bigcap A_n$ then $\exists n$ st.

$\operatorname{diam} A_n < |b - a|$ $\lightning$.

Pugh 2.12. (p_n). $f: \mathbb{N} \to \mathbb{N}$ reor. $q_k = p_{f(k)}$.

if f inj, $\lim p_n = \lim q_n$.

if f surj, no. e.g. f: $1 \mapsto 1$, $2 \mapsto 2$, $3 \mapsto 2$, $4 \mapsto 3$, $5 \rightrightarrows$

$(p_n) = (1, 1 - \frac{1}{2}, 1 - \frac{1}{3}, -\frac{1}{4}, 1 - \frac{1}{5}, -\frac{1}{n})$.

limit does not exc.

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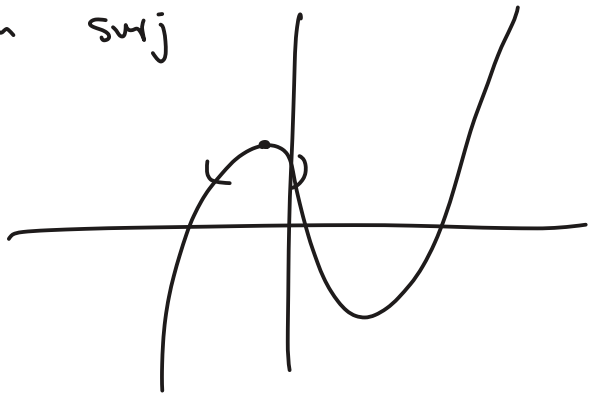
Pugh 28.d
cts non open surj



f) no, $z \mapsto z^n$ $n>1$
cts, open, surj, not inj.

e) wts injective. if not, $a \neq b \longmapsto f(a) = f(b)$.
$[a,b] \subset \mathbb{R}$ cpct $\Rightarrow$ min x_m, max x_M. if
x_m and $x_M \in \{a,b\}$ then f const on (a,b), not open.
So WLOG $x_m \in (a,b)$. $f((a,b)) = [m, M)$ or $[m, M]$. ξ.

Pugh 47. a) no, $(0,1)$ homeo $\mathbb{R}$ but

b)

but

c) A, B cpct, crcted homeom. Then $\mathbb{R}^2 \setminus A$, $\mathbb{R}^2 \setminus B$ homeomorphic.
intuition: complements are disjnt $\cup$ opens

TBD: $[74, 95] \cup [96, 100] \cup [102, 114] \cup [116, 151]$.
$[P1, P15]$.

$f: K \to \mathbb{R}$ cts, $\epsilon > 0 \Rightarrow \exists M$ st. $|f(x) - f(y)| \leq M|x-y| + \epsilon$.
$K \subset \mathbb{R}^m$

$f: I^2 \to \mathbb{R}$ cts. $g(x) = \max_y f(x,y)$ cts.

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Pugh P2.1. $f: \mathbb{R}^m \to \mathbb{R}$ st. (1) K cpct $\subset \mathbb{R}^m \Rightarrow f(K)$ cpct

(2) $\underset{\mathbb{R}^m}{\underset{\cap}{K_1}} \supset K_2 \supset \cdots$ cpct $\Rightarrow f(\bigcap K_n) = \bigcap f(K_n)$.

Show f cts.

Proof. $x_n \longrightarrow x$ in $\mathbb{R}^m$. wts $f(x_n) \to f(x)$.

$K_n := \{x_n, x_{n+1}, \cdots\}$. cpct as any sequence in K_n, i.e. subseq of (x_n), has cgent subseq.

$K_1 \supset K_2 \supset \cdots$ so $f(\bigcap K_n) = \bigcap f(K_n)$.

Claim: $\bigcap K_n = \{x\}$. Indeed $\operatorname{diam} K_n \longrightarrow 0$

as $\forall \epsilon > 0, \exists N$ st. $m, n > N \Rightarrow |x_m - x_n| < \epsilon$

So $\operatorname{diam} K_N = \sup_{m,n \geq N} |x_m - x_n| \leq \epsilon$.

So $\bigcap f(K_n) = \{f(x)\}$.

Claim: every cgent subseq of $f(x_n)$ cges to $f(x)$.

Then every subseq of $f(x_n)$ has a cgent subseq

Since $\{f(x_n)\} = f(K_1)$ cpct.

this cges to $f(x)$

So $f(x_n) \longrightarrow f(x)$.

$f(x_{n_k}) \longrightarrow y$. we show $y = f(x)$ by showing

$y \in \bigcap f(K_n)$

$\forall n \in \mathbb{N}$, y is limit pt of $f(K_n)$

e.g take $n_k > n$,

$(f(x_{n_k}), f(x_{n_{k+1}}) \cdots) \to y$.

Since $f(K_n)$ cpct $\Rightarrow$ closed, $y \in f(K_n)$.

So $y = f(x)$.

Pugh P 2.3. $f: \mathbb{R}^2 \to \mathbb{R}$ st. $y \mapsto f(x_0, y)$ cts $\forall x_0 \in \mathbb{R}$ but f not cts.

$x \mapsto f(x, y_0)$ cts $\forall y_0 \in \mathbb{R}$

$f(x,y) = \{$

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Young $e^{tX+(1-t)Y} \le te^X + (1-t)e^Y$

$= e^{tX} e^{(1-t)Y}$

Let $t = \frac{1}{p}$, so $1-t = \frac{1}{q}$

Let X be s.t. $e^{tX} = x$, i.e. $X = p \log x$

Y s.t. $e^{(1-t)Y} = y$ $\quad Y = q \log y$.

Hölder $a \in \ell^p, b \in \ell^q$. $\sum |a_n||b_n| \le \|a\|_p \|b\|_q$.

WLOG $\|a\|_p = \|b\|_q = 1$.

Young $\Rightarrow$ $\sum |a_n||b_n| \le \sum \frac{1}{p}|a_n|^p + \frac{1}{q}|b_n|^q$

$= \frac{1}{p} + \frac{1}{q} = 1$.

Minkowski $\|a+b\|_p \le \|a\|_p + \|b\|_p$

$$\sum |a_n+b_n|^p \le \sum |a_n+b_n|^{p-1}(|a_n|+|b_n|)$$

$$\le (\|a\|_p + \|b_p\|) \left(\sum |a_n+b_n|^{q(p-1)}\right)^{\frac{1}{q}}$$

$$= (\|a\|_p + \|b\|_p) \left(\sum |a_n+b_n|^p\right)^{1-\frac{1}{p}}$$

$q = \frac{p}{p-1}$

$\frac{1}{q} = 1 - \frac{1}{p}$

$$\left(\sum |a_n+b_n|^p\right)^{\frac{1}{p}} \le \|a\|_p + \|b\|_p.$$

$p < q$. $\quad s \in \left(\frac{1}{q}, \frac{1}{p}\right)$

$\frac{1}{n^s} \in \ell^q \setminus \ell^p$. $\quad \left(\sum \frac{1}{n^{sp}}\right)^{\frac{1}{p}}$ $\quad sp > 1$

div.

$\left(\sum \frac{1}{n^{sq}}\right)^{\frac{1}{q}}$ cges.

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